Tirosh 2016 IDH-mut oligodendroglioma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 6 tumours intermix per-neighbourhood (iLISI 1.02→4.30 of 6 under Harmony θ=10) on ONE continuous manifold (single kNN component, 0 islands removed) — no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

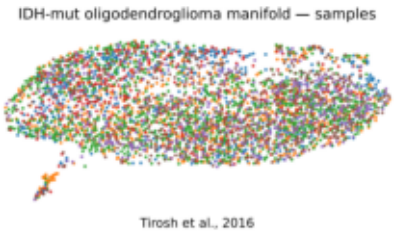
Gast/pSom co-vary (r=+0.38) and BOTH oppose Crest (−0.85 / −0.59): a single developmental axis with Crest an inverse terminal gradient. Non-branching case — GO. Crest enriches the terminus, not a Gast arm.

Gate 3 — cross-repression → NA

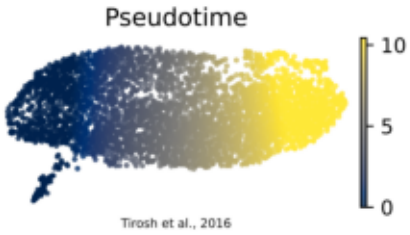
No bifurcation — scFates curve is robustly fold-free (2 tips, 0 forks, self-approach 2.99 across Nodes 15–30).
Hub cross-repression is not defined for a linear axis — Gate 3 not tested.

STEP FIGURES

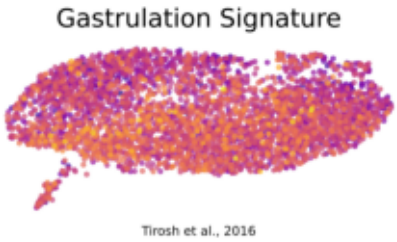
Manifold — samples



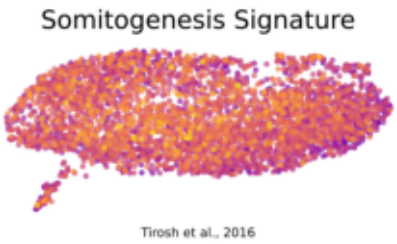
Pseudotime



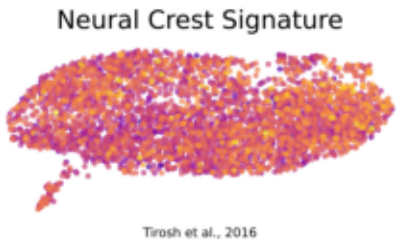
Gastrulation



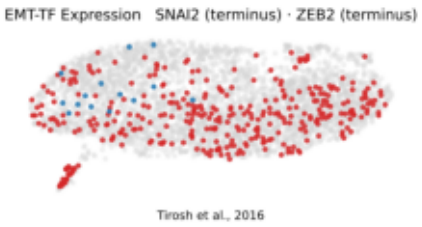
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-mutant oligodendroglioma (Tirosh et al., 2016), SmartSeq2 (TPM) — 4,089 malignant cells across 6/6 samples. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic-program pattern reproduces as the non-branching case: one continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Geometry Linear-pSom/Crest (pSom μ=+0.16 globally strongest; Gast −0.04, Crest −0.20). EMT-TFs SNAI2 · ZEB2 (both rise to the Crest terminus).

Parameter choices

Min-cell threshold	≥100 → 6/6 samples (4,089 cells); counts descend smoothly 1185→428	**Genes / embedding**	5,800 overdispersed genes; n_pcs=20; Harmony θ=10 on sample
Islands removed	0 (single kNN component; continuous manifold)	**Geometry**	non-branching curve (N=20, λ=0.3, μ=100): 2 tips / 0 forks, fold-free
Root node	curve node 2 = Gast/pSom-high, Crest-low origin	**EMT-TFs**	ZEB1 (82%), ZEB2 (83%), PRRX1 (38%), SNAI2 (3%) rise to Crest; display SNAI2 · ZEB2